

Satellite Design & Integration

DAY 3



Satellite Set Up

Let's start building the 1U Cubesat!

Follow the guide:

- Read carefully and understand each step before proceeding.
- Strictly adhere to the guide's sequence; do not skip or advance independently.

Do not power subsystems until instructed:

- Avoid connecting or powering subsystems prematurely to prevent conflicts. Keep the RBF plugged in.

Use the screwdriver set correctly:

- Use the right size of the screwdriver inserts for each specific screw.
- Avoid over-tightening screws; tighten securely without excessive force to prevent damage.

Satellite Set-up - EPS

First of all: find the RBF (Remove Before Flight) jack and plug it in the EPS



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Satellite Set-up - EPS



Satellite Set-up - EPS

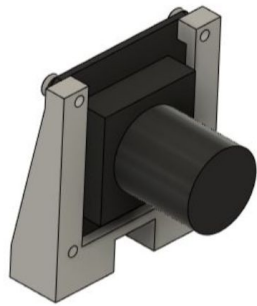
Let's attach the battery to the EPS using one or two pieces of double-sided tape and then plug the battery into its connector.



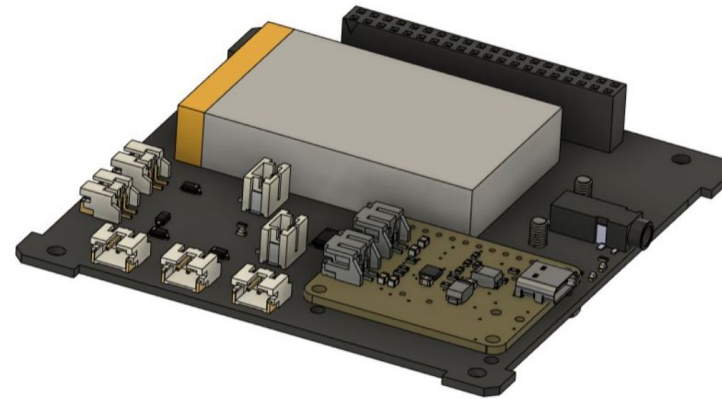
Satellite Set-up - EPS



2xM2 x 4

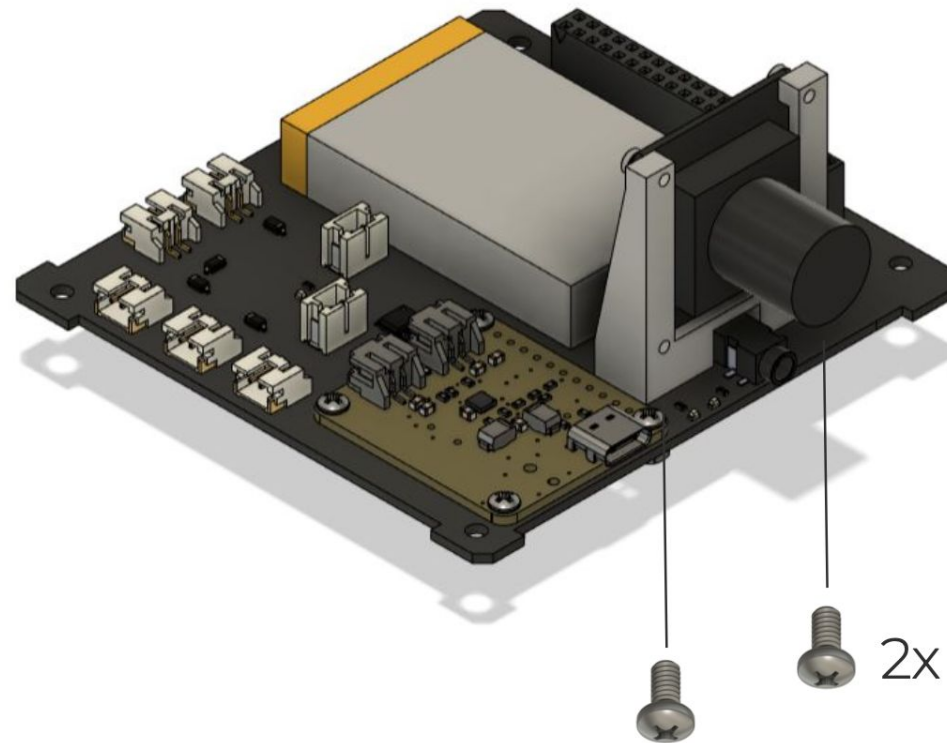


1x Camera



1x EPS Board

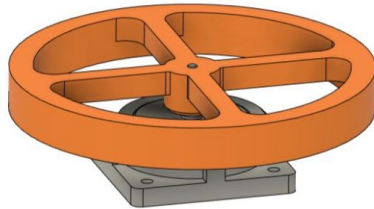
Satellite Set-up - EPS



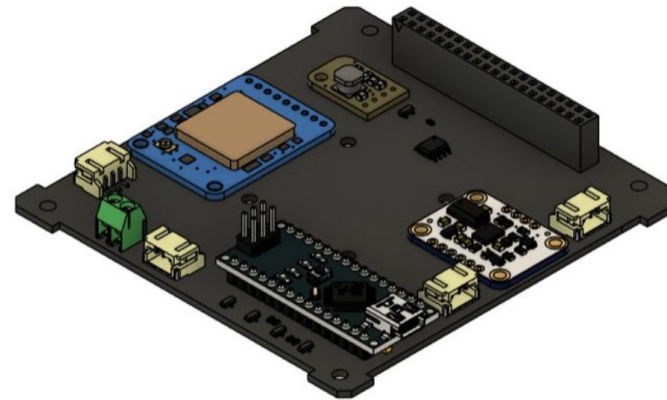
Satellite Set-up - ADCS



4xM2x4



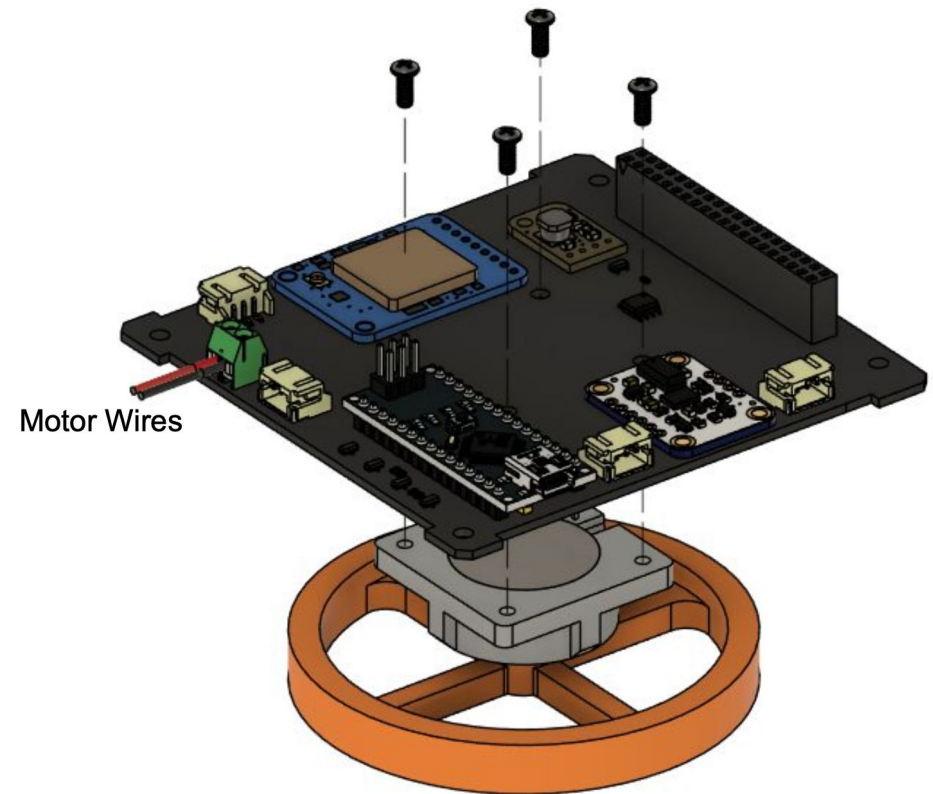
1x Reaction Wheel



1x ADCS Board

Satellite Set-up

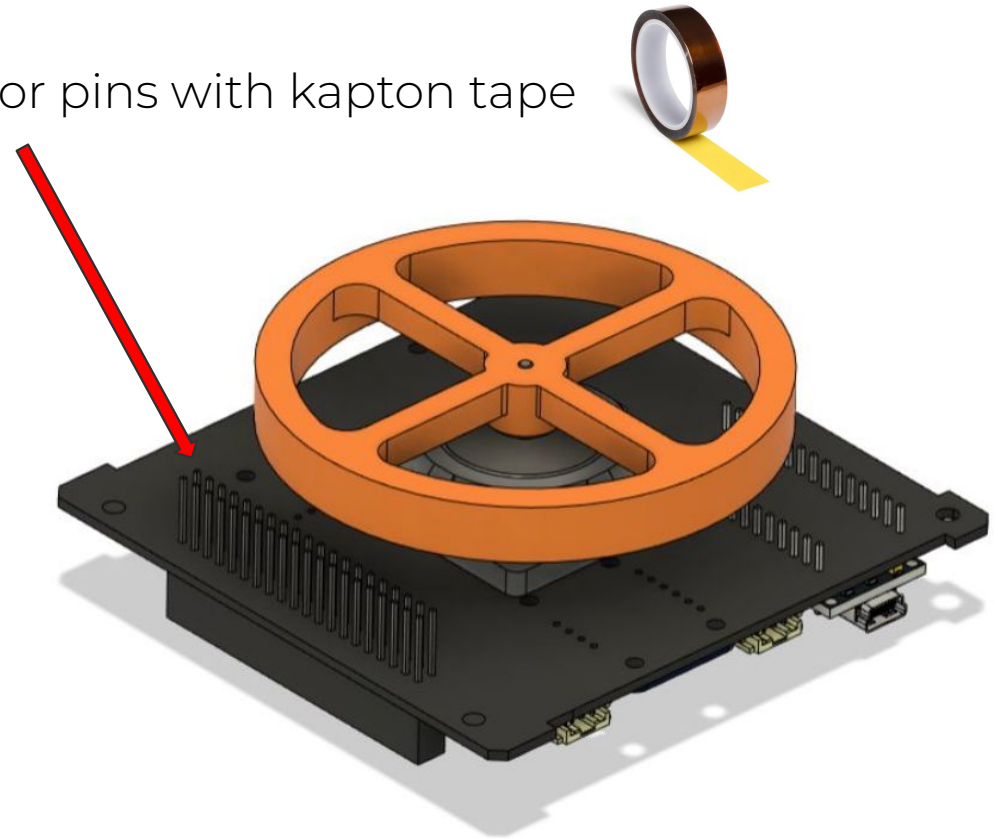
Be careful with the motor wires, unroll them from the motor case and plug them into the connector.



Satellite Set-up - ADCS

Let's insulate the PC-104 connector pins with kapton tape

Use the same tape to secure motor wires to the board so that they don't hit the wheel or get close to exposed pins.



Satellite Set-up - Bottom side



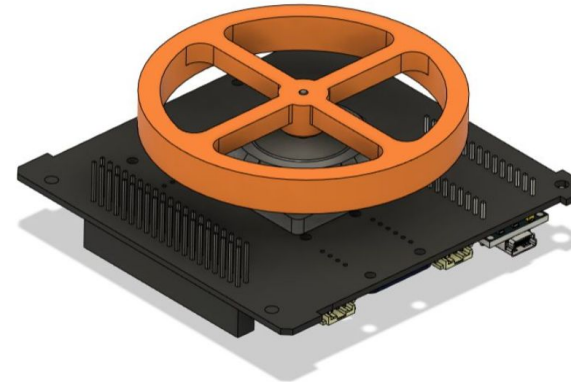
4xCBSTSR M3x10



4x20mm Spacers



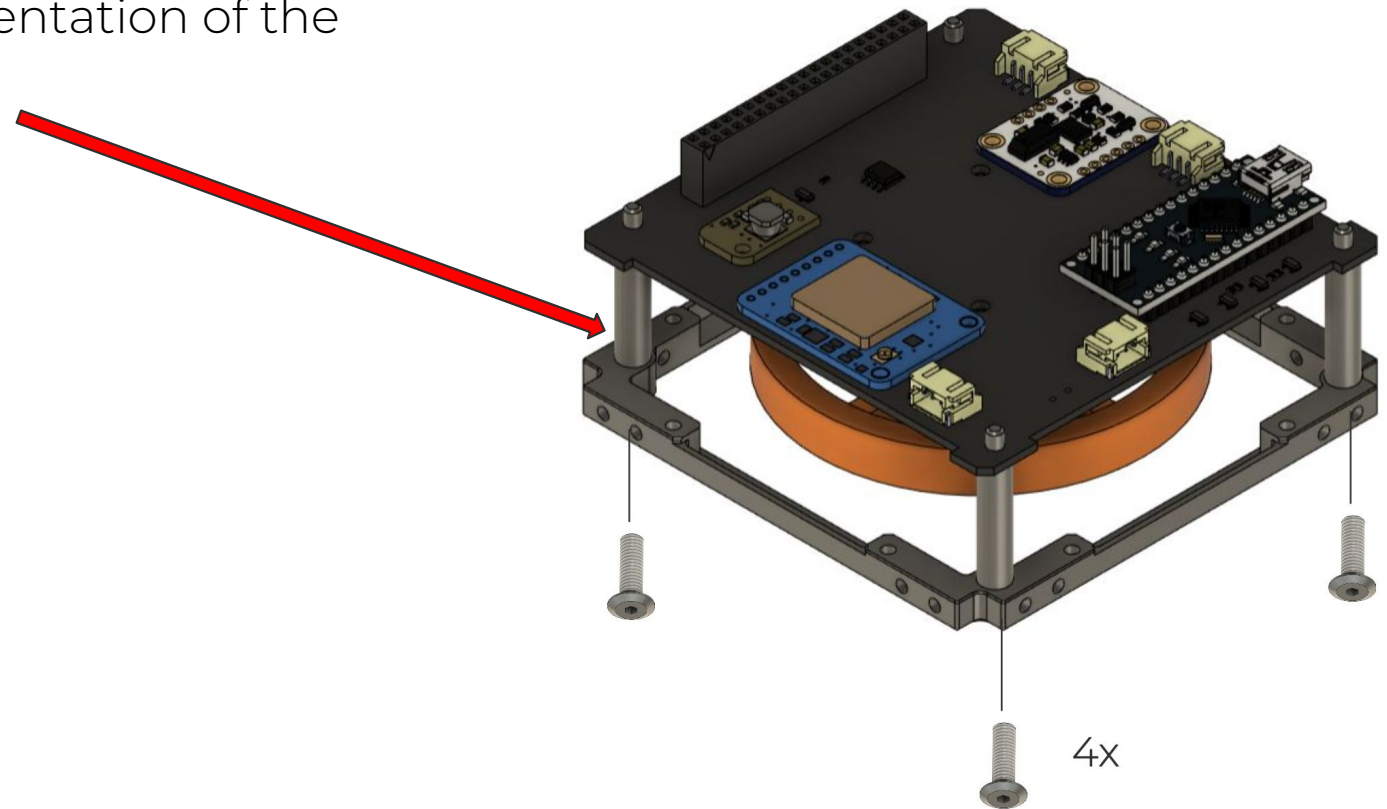
1xVRS006M



1x ADCS Board

Satellite Set-up - Bottom side

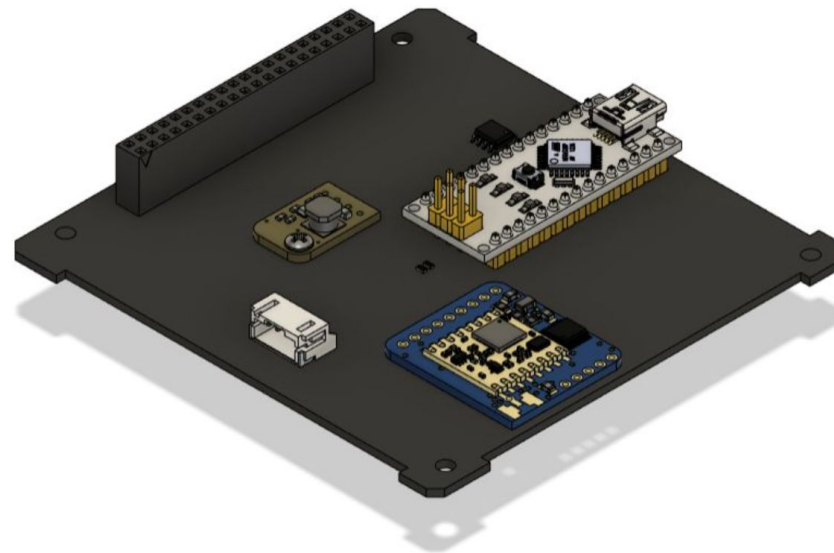
Look at the orientation of the bottom frame!



Satellite Set-up - OBC and ADCS Stack

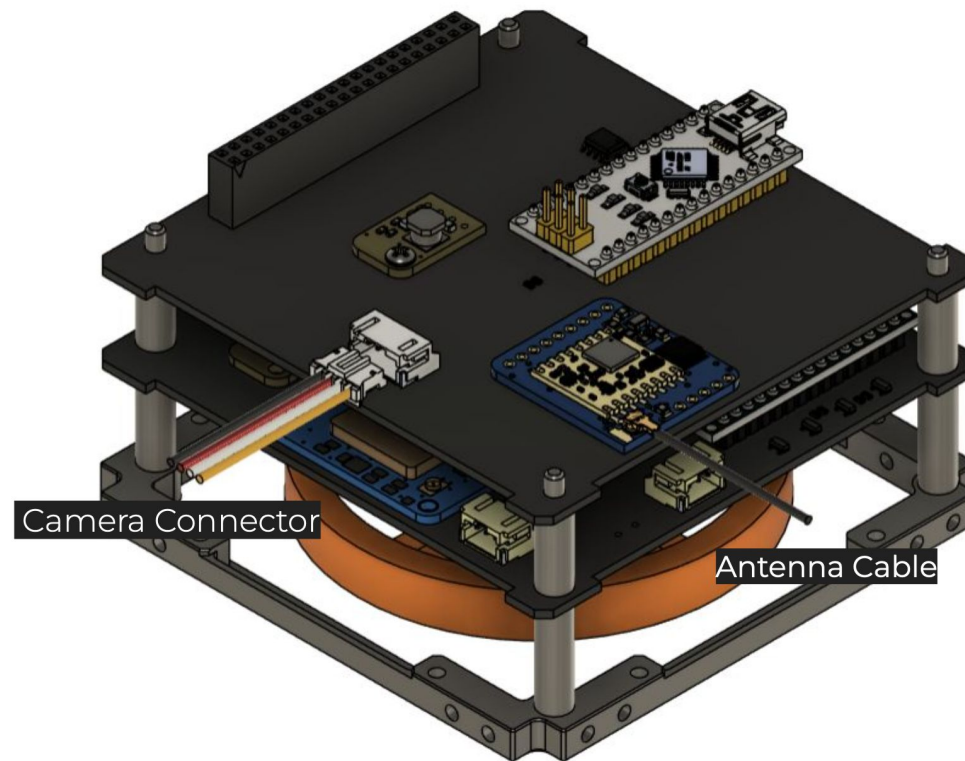


4x15.3mm Spacers



1x OBC Board

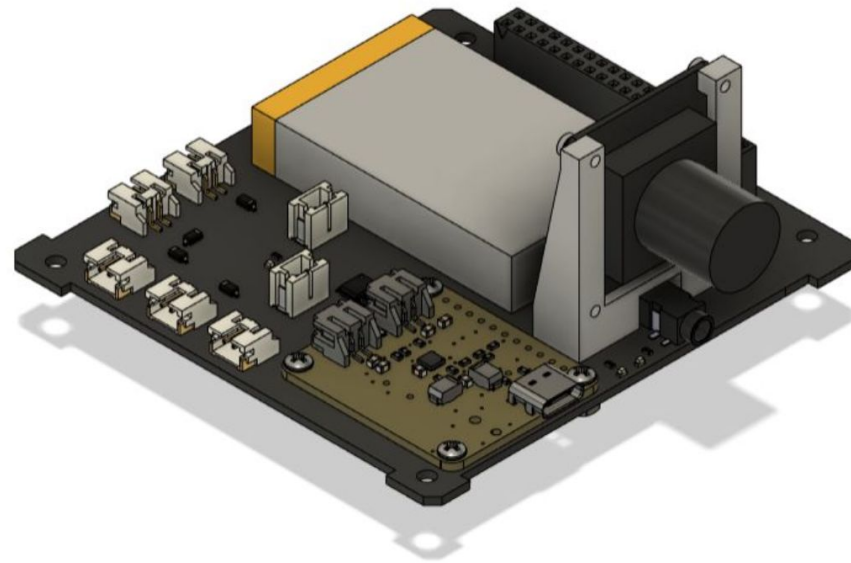
Satellite Set-up - OBC and ADCS Stack



Satellite Set-up - Preparing the Full Stack



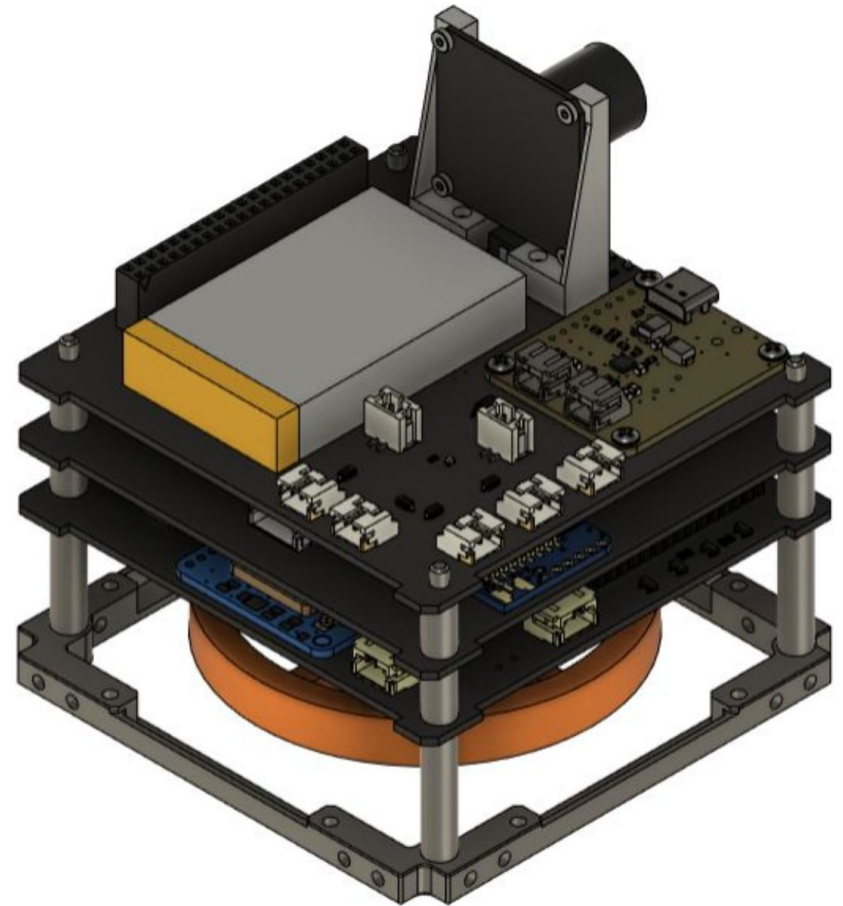
4x15.3mm Spacers



1x EPS Board +
Camera

Satellite Set-up - Full Stack

Plug the camera connector to the OBC BEFORE placing the EPS on top of it!



Satellite Set-up - Top Frame

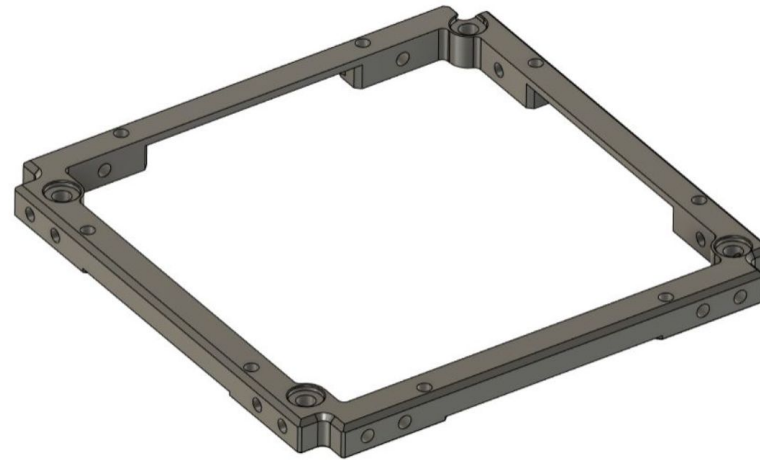
Build the top frame of the 1U in a similar way as for the bottom frame



4x30.25 Spacers

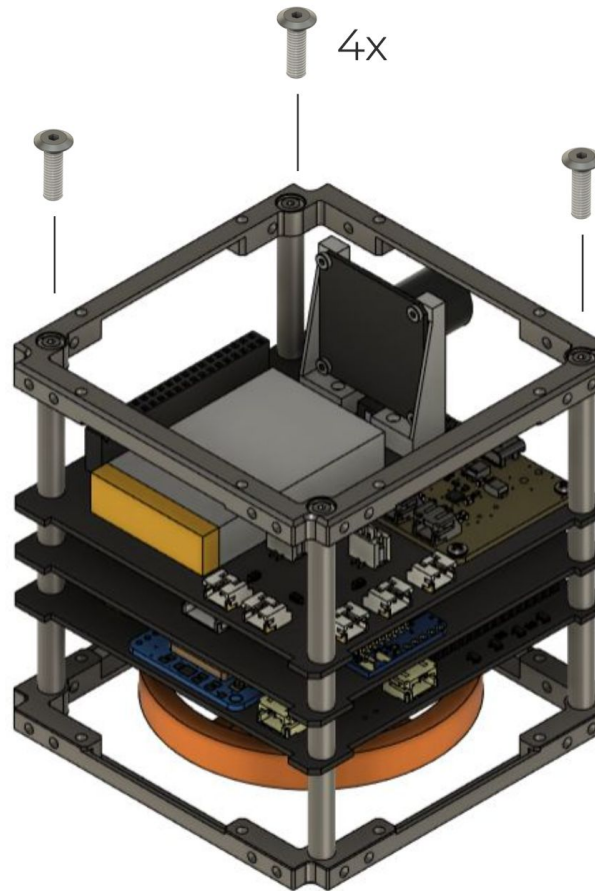


4xCBSTSR M3x10



1xVRS006

Satellite Set-up - Top Frame



Satellite Set-up - Side Structures



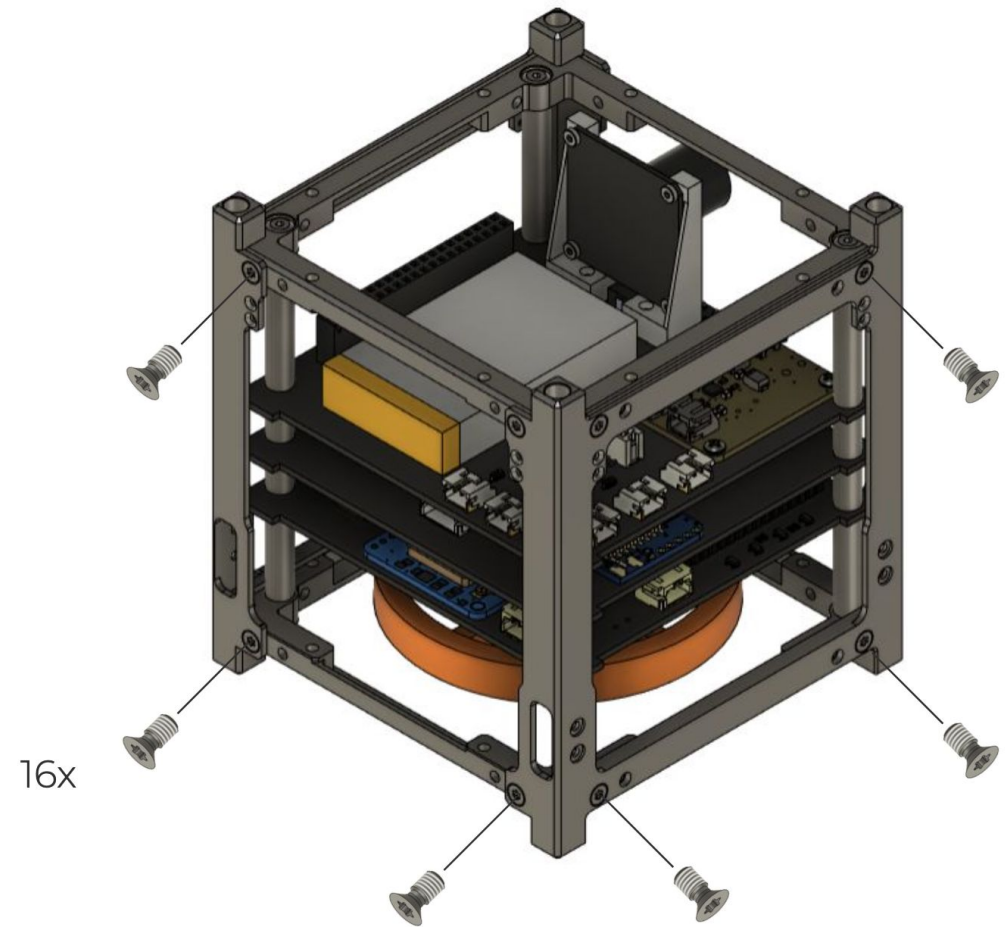
16x TORX M2.5x5



2xVRS001

Satellite Set-up - Side Structures

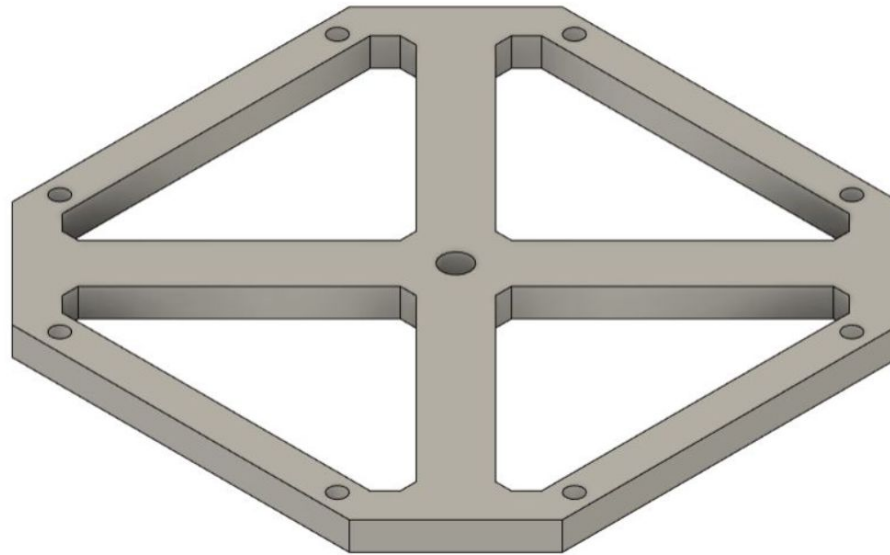
Pay attention to the mounting direction of the side structures



Satellite Set-up - Bottom Guard

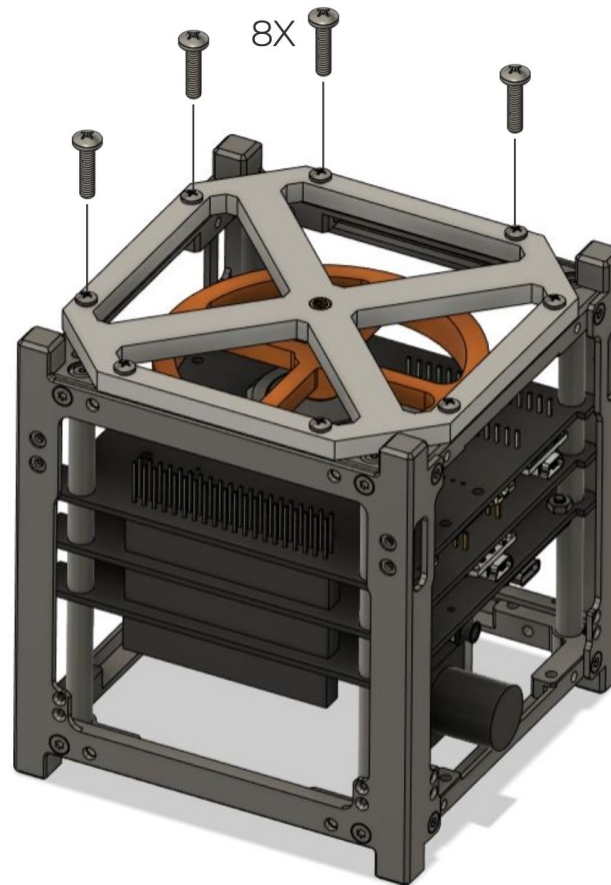


8xM2.5x10



1x Support

Satellite Set-up - Bottom Guard



Satellite Set-up - Boards Check

Before continuing with the satellite mechanical set-up, let's make sure our boards can be programmed!

Remember:

- **Arduino Nano** - OBC & Ground Station
- **Arduino Nano Every** - ADCS

Satellite Set-up - Boards Check

Hardware side:

- Connect the USB C to the EPS and plug it in your PC.
- Now remove the RBF.

Software side:

- Open a new sketch in Arduino IDE - do not write any code in it

```
1 void setup() {  
2   // put your setup code here, to run once:  
3  
4 }  
5  
6 void loop() {  
7   // put your main code here, to run repeatedly:  
8  
9 }  
10
```


Satellite Set-up - Boards Check

Software side:

- Open a new sketch in Arduino IDE - do not write any code in it
- Connect and upload the empty code on the OBC first.
- Then, disconnect the OBC cable.
- Connect and upload the empty code on the ADCS.

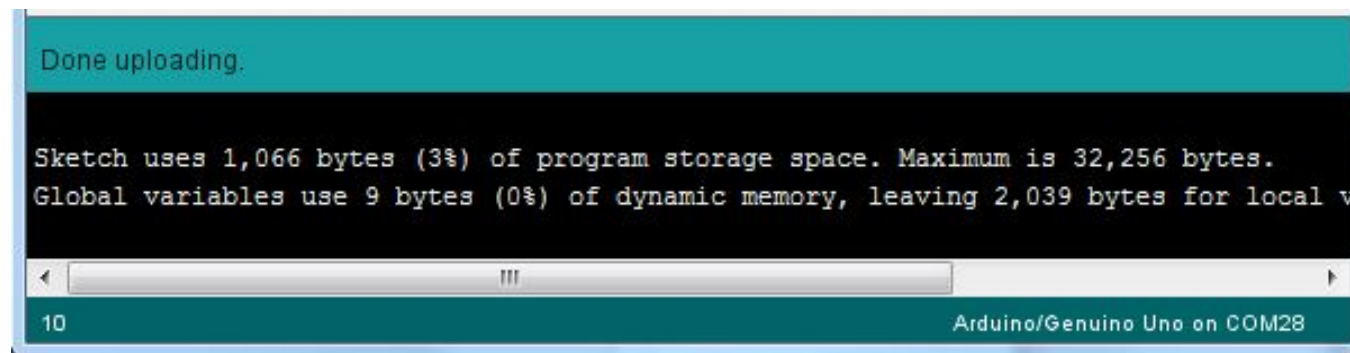
```
1 void setup() {  
2   // put your setup code here, to run once:  
3  
4 }  
5  
6 void loop() {  
7   // put your main code here, to run repeatedly:  
8  
9 }  
10
```

Satellite Set-up - Boards Check

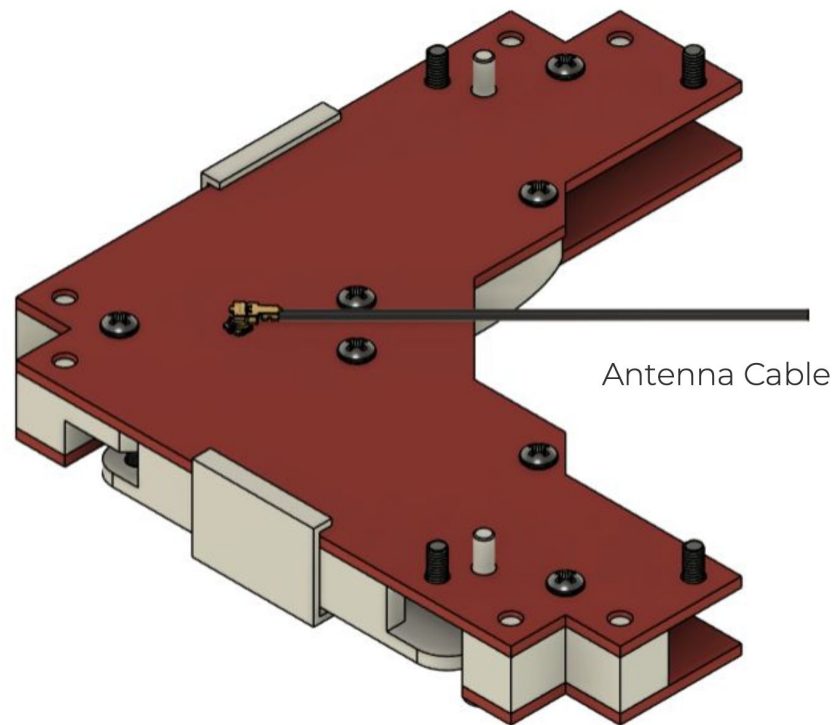
Hardware side:

- Plug the RBF back in.
- Remove the remaining Arduino cable.
- Remove the USB C cable from the EPS.

Now we know that both OBC and the ADCS can be programmed!



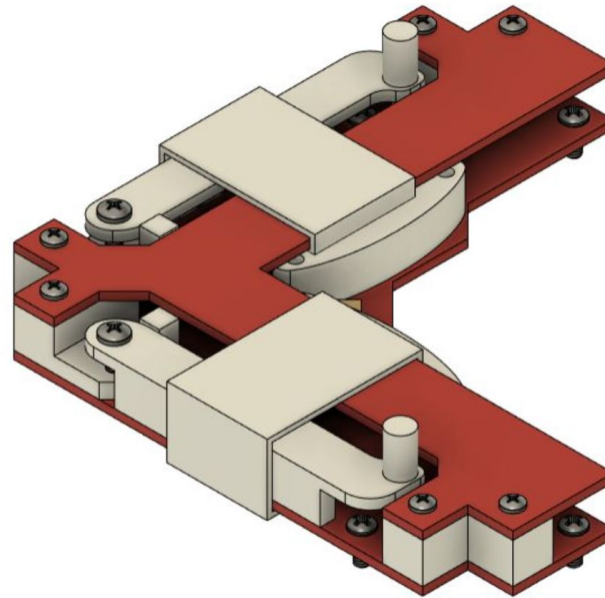
Satellite Set-up - Antenna OBC Connection



Satellite Set-up - Antenna



2xM2.5x6

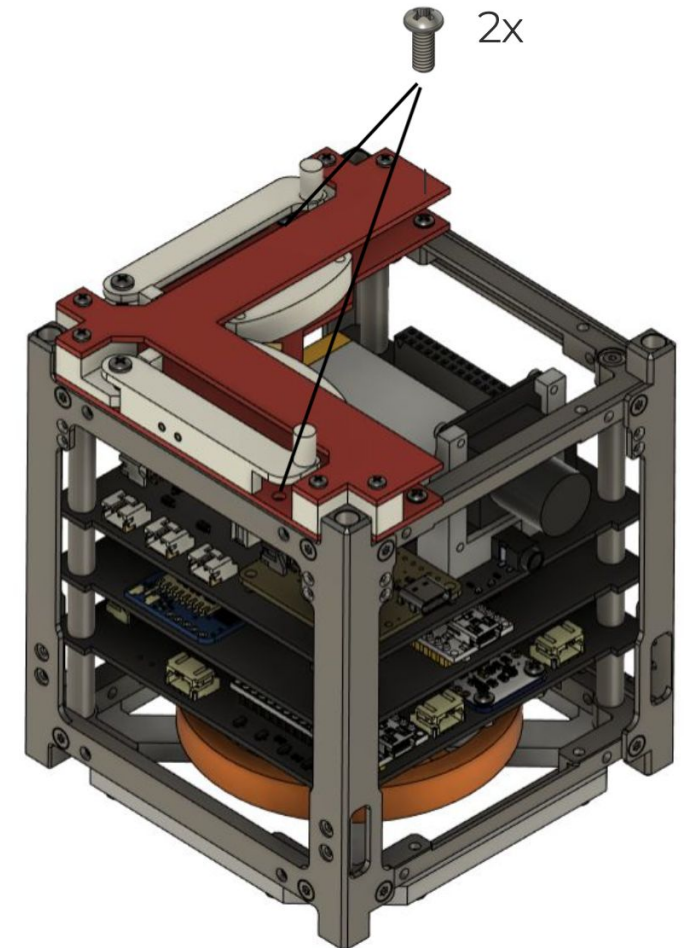


1xDeployable Antenna

Satellite Set-up - Antenna

To insert the M2.5x6, open manually
deploy the antennas!

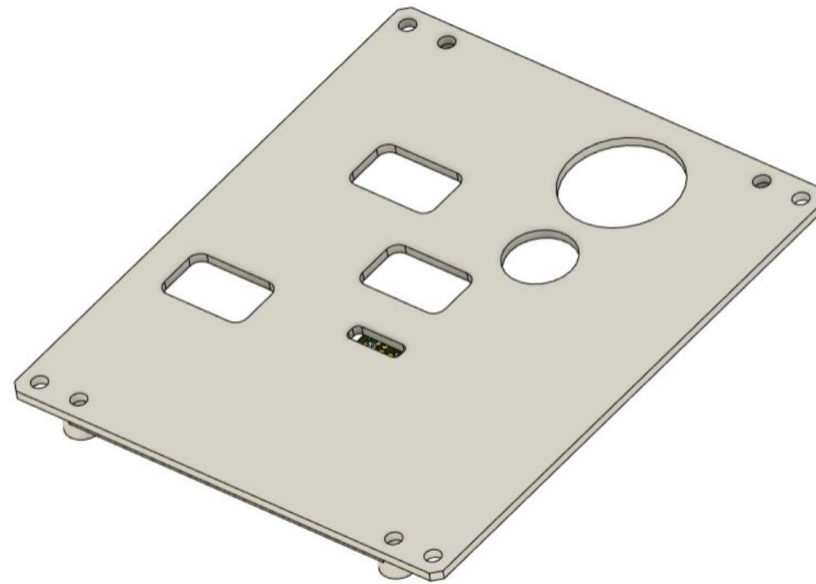
Once done, close them back with the
pins.



Satellite Set-up - Side Panel



4xM2.5x10

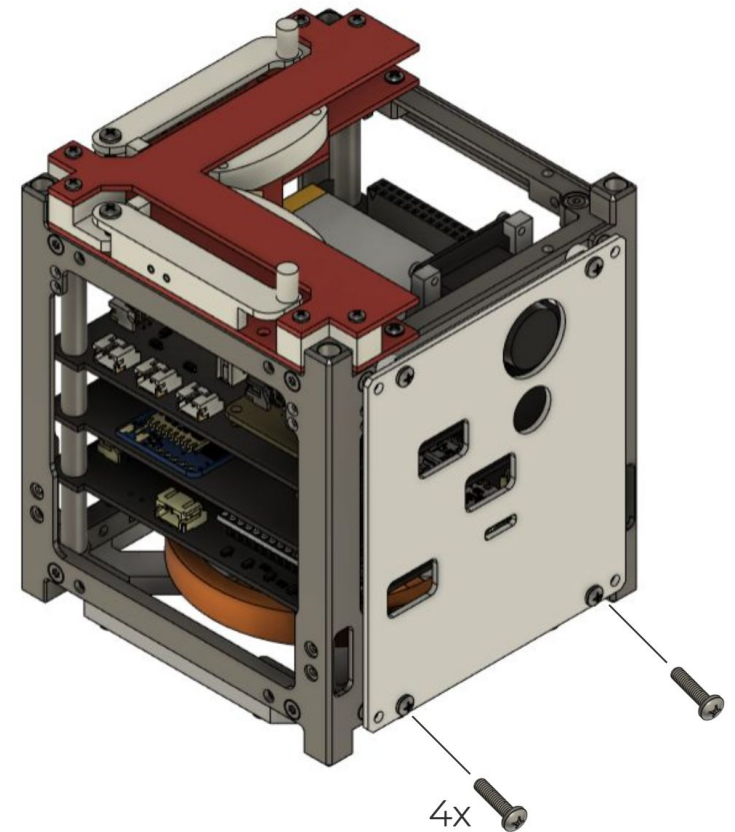


1x Side Panel

Satellite Set-up - Side Panel

To correctly place the panel, we are going to need back spacers!

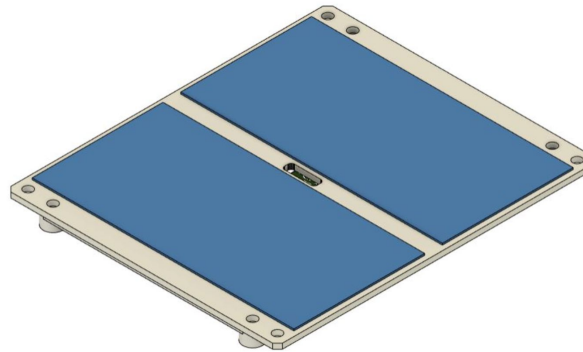
After placing the side panel, take some kapton and label EPS, ADCS, RBF and OBC with a marker.



Satellite Set-up - Solar Panels

Solar panel cells are extremely fragile and tend to be broken easily if touched.

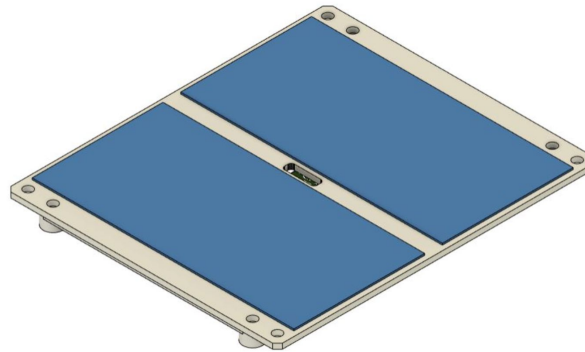
- Avoid touching them directly with your fingers.
- Be careful not to hit them with tools.
- Most importantly, do not place the faces with solar panels directly on the table or onto other components.



Satellite Set-up - Solar Panels

Before mounting the solar panels onto the satellite, it's essential to ensure their functionality.

- Begin by using a multimeter to measure voltage output, checking each panel individually using the torchlight.
- Additionally, inspect the panels to identify any physical defects or anomalies.



Solar Panels

A solar panel, or photovoltaic panel, is a device that converts sunlight into electricity.

The basic unit of a solar panel is the solar cell which is made of semiconductor materials.

It is known as “photovoltaic effect” the process through which the sunlight generates electric current.

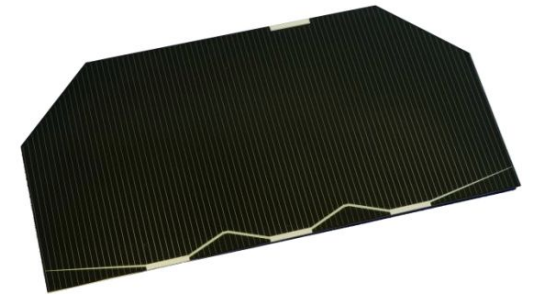


Space Grade Solar Panels

Conventional solar cells used for civilian applications typically utilize a double-junction design. In contrast, triple-junction solar cells, preferred in space missions, prioritize heightened efficiency over concerns of cost and weight.

Engineered to effectively capture a wide spectrum of solar radiation, these cells excel in the challenging conditions of space with limited sunlight. Triple-junction cells significantly enhance energy absorption and power generation.

Their durability in harsh space environments, including exposure to radiation and extreme temperatures, positions them as an optimal choice for powering satellites, space probes, and scientific instruments.



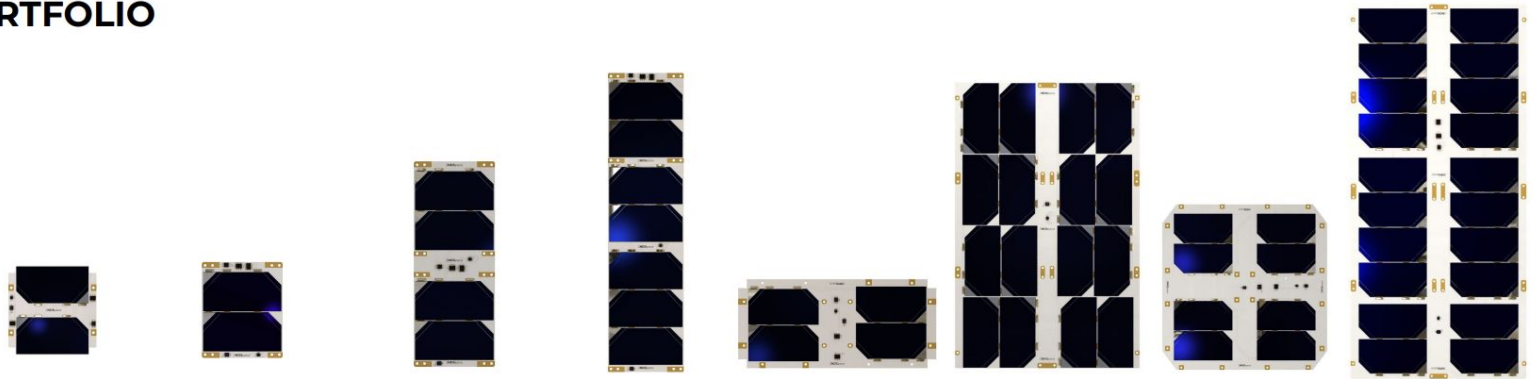
Space Grade Solar Panels

If we consider a double row of solar cells on each face of a CubeSat, the number of cells may be distributed as follows:

- **1U CubeSat:** A 1U CubeSat with a double row of solar cells might have around 6 to 12 solar cells.
- **2U CubeSat:** For a 2U CubeSat, a double row could result in approximately 12 to 24 solar cells.
- **3U CubeSat:** A double row on a 3U CubeSat might feature around 18 to 36 solar cells.
- **6U CubeSat:** On a 6U CubeSat, a double row configuration could include approximately 36 to 60 solar cells.
- **12U CubeSat:** For a 12U CubeSat, with a double row, you might find around 60 to 120 solar cells (deployables).

2NDSpace

PRODUCT PORTFOLIO



	CORE-01Z	CORE-01	CORE-02	CORE-03	CORE-06Z	CORE-06	CORE-12Z	CORE-16
Peak Power [W]	2.5	5	5	8,75	5	20	10	25
Suggested for	1U/2U/3U	2U/3U	2U/3U	3U/6U	6U	6U/12U	12U/16U	16U
Form Factor	1x1	1x1	1x2	1x3	2x1	2x3	2x2	2x4
Weight [g]	38	35	75	118	91	289	195	387

QUALIFICATION AND ACCEPTANCE TEST

	Functional/AIT	Electrical Test/Flash	Vibration test	Mechanical Shock	TVAC Test
Qualification Test	✓	✓	NASA GEVS: GSFC-STD-7000A ESA ECSS-E-ST-10-03C	NASA GEVS: GSFC-STD-7000A ESA ECSS-E-ST-10-03C	NASA GEVS: GSFC-STD-7000A ESA ECSS-E-ST-10-03C
Acceptance Test	✓	✓			

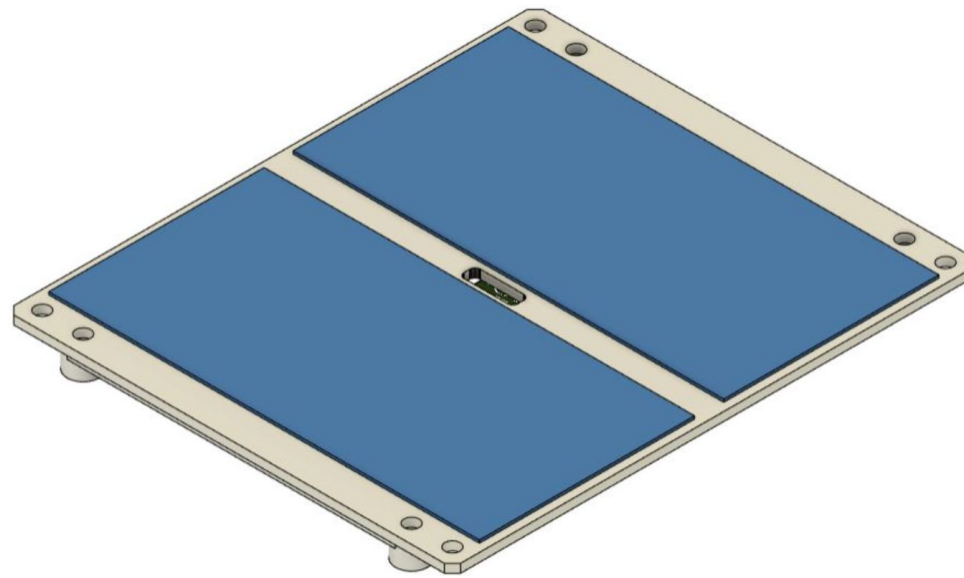
Space Grade Solar Panels

Let's have a look to a real 1U solar panel!

Satellite Set-up - Solar Panels



12xM2.5x10



3x Solar Panels

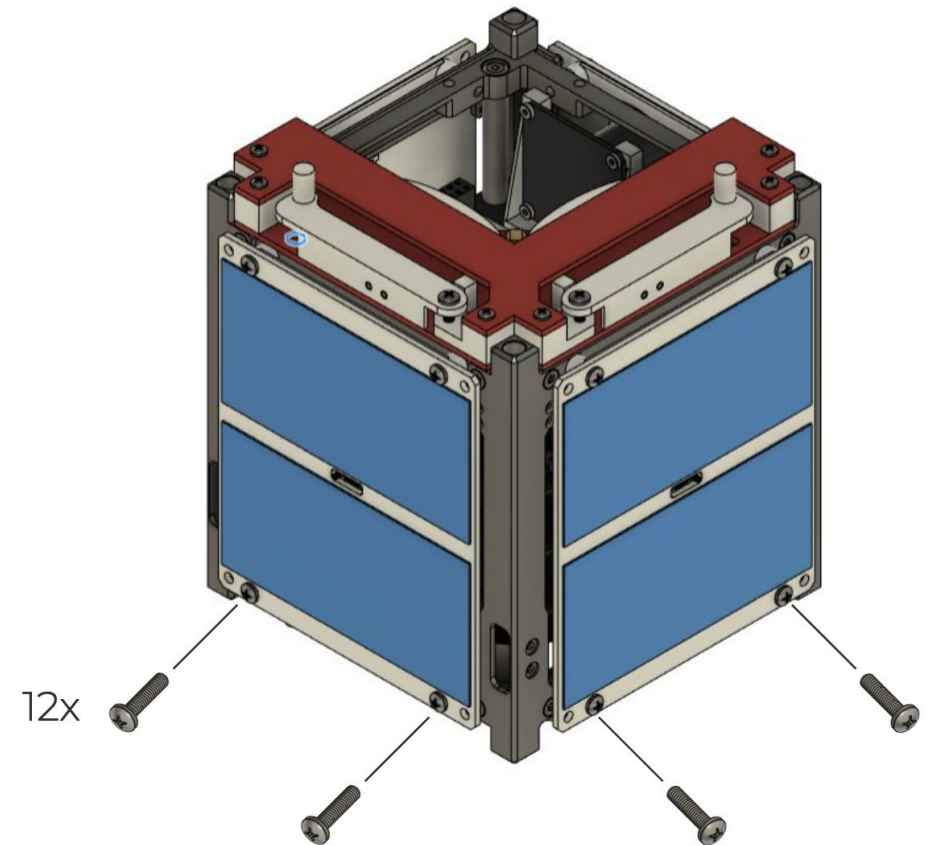
Satellite Set-up - Solar Panels

Before placing the solar panels we need to:

- Connect the solar cells outputs to the EPS.
- Connect the sun sensors to the ADCS.

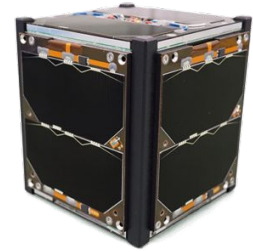
Refer to the EPS and ADCS schematics to understand where to connect each cable.

Use the spacers behind the panels to place them on the satellite.



Satellite Set-up - Full Satellite

Congratulations, you built the entire 1U Cubesat!



Satellite Set-up - Discussion

Let's talk about satellite integration, tests and launch.

Hardware integration experience:

- What did you find to be the most difficult part in building the satellite?
- Which components are the most challenging to handle?

Let's have a look at where real satellites are built.

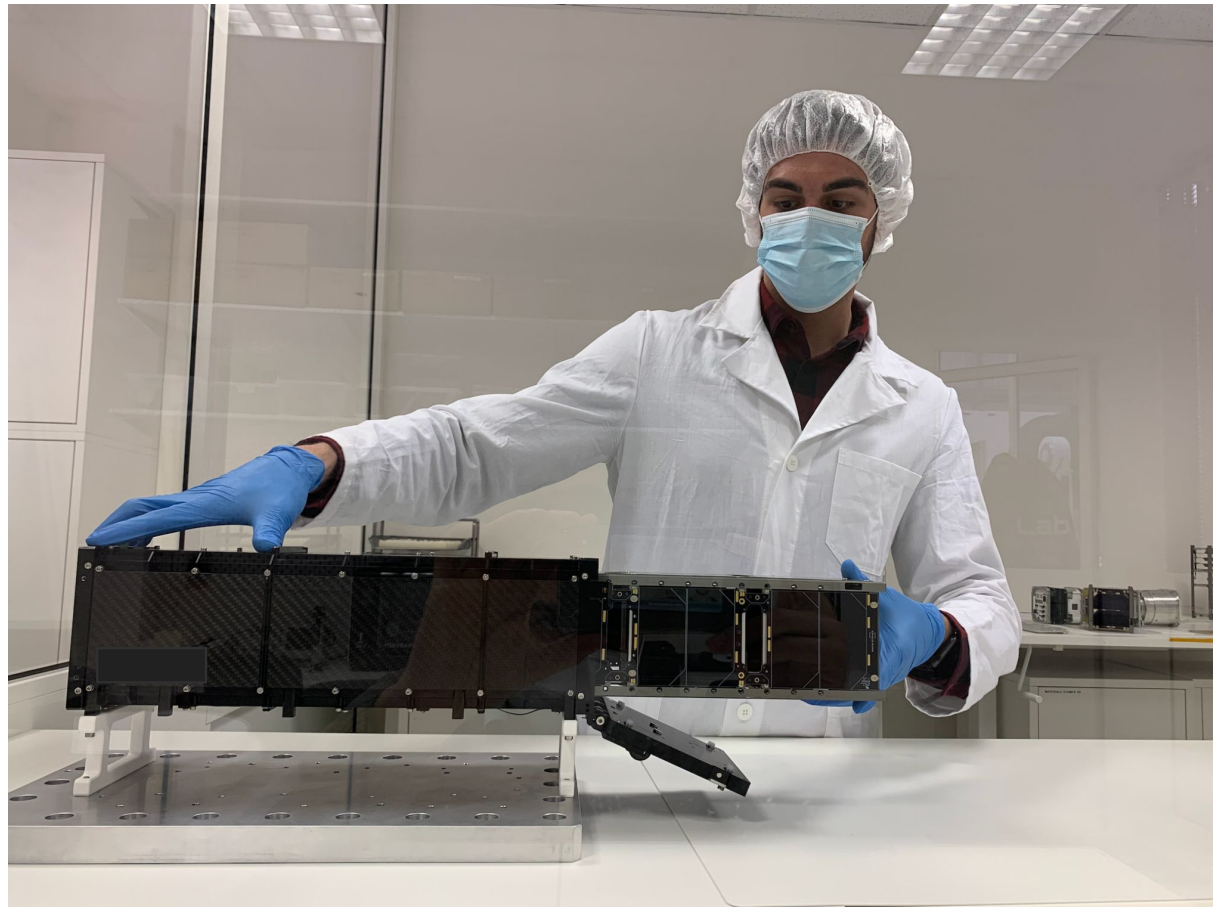
Satellite Set-up - Discussion

Real satellites are assembled and tested in **cleanrooms**:

A cleanroom is a controlled environment that is designed to minimize contamination and maintain a specific level of cleanliness.



Satellite Set-up - Discussion



Satellite Set-up - Discussion

Key aspects of a cleanroom:

- *Controlled Particle Count:* Cleanrooms control the concentration of airborne particles per cubic meter.
- *Temperature and Humidity Control:* Cleanrooms maintain precise temperature and humidity levels.
- *Airflow Control:* Airflow is carefully filtered to minimize the introduction of particles.
- *Positive Pressure:* Cleanrooms are often maintained at a slightly higher air pressure than the surrounding areas to prevent the entry of contaminants from outside.
- *Personnel Protective Equipment (PPE):* Individuals entering a cleanroom typically wear specific clothing, including coveralls, gloves, masks, and sometimes even full-body suits.
- *Strict Protocols:* Cleanroom protocols, including entry and exit procedures.

Satellite Set-up - Discussion

ESD stands for Electrostatic Discharge, and ESD devices are designed to protect electronic components and devices from damage caused by electrostatic discharge.

ESD Wrist Straps: Worn by personnel to ground themselves and prevent the build-up of static charge on their bodies. The wrist strap is typically connected to a grounding point, therefore limiting the range of movement of the operator.



ESD Mats: Placed on work surfaces to provide a grounded surface, preventing the accumulation of static charge on the workbench.

Satellite Set-up - Discussion

Satellite components, like spacers and screws, are mainly made of metal. CubeSats have limited space filled with cables, making it challenging to work inside.

Accidentally, bolts and screws can fall during assembly, posing risks like short circuits.

Retrieving them is vital to avoid hardware damage during the launch's intense vibrations.

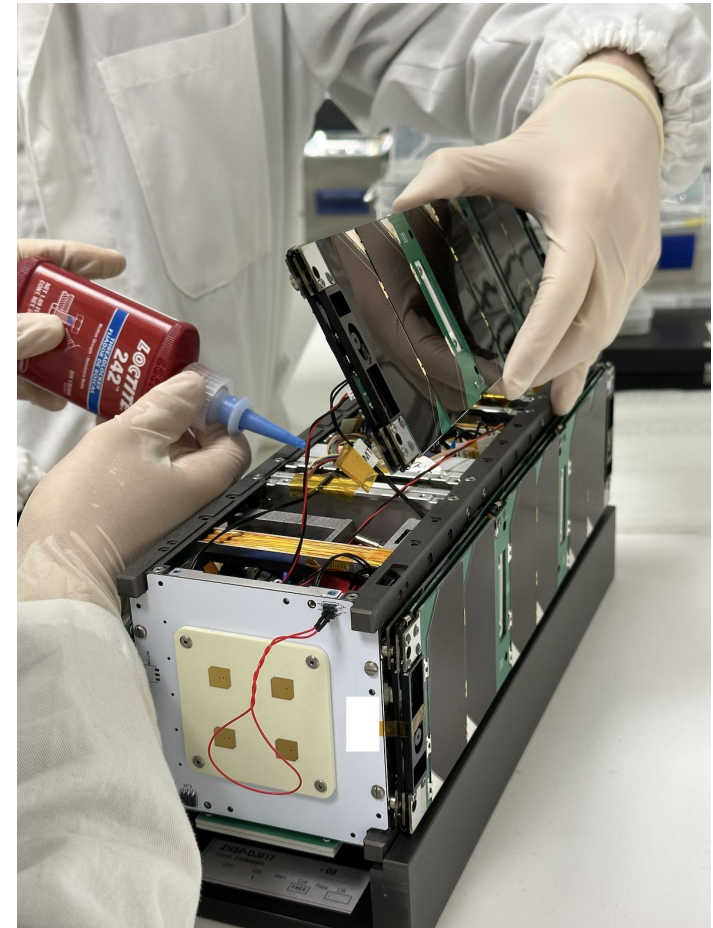
Though it's hard to confirm if any were left inside, there are concerns that some missions, even famous ones, might have failed due to overlooked components like forgotten screws.



Satellite Set-up - Discussion

Even screws placed correctly can loosen due to satellite vibrations during tests or rocket launches.

As a result, they are not simply screwed in; a special glue is applied before insertion, and the head is then sealed with a specialized epoxy resin known as araldite.



Satellite Set-up - Discussion

Cubesats undergo a series of rigorous tests before launch to ensure their functionality and reliability in space. Generally tests include the following:

- *Vibration Testing:* Cubesats are subjected to simulated launch vibrations to mimic the conditions experienced during rocket liftoff. This test helps identify any structural weaknesses or potential loose components that could result from the intense vibrations.
- *Thermal Vacuum Testing:* Cubesats are exposed to extreme temperatures and vacuum conditions to replicate the harsh environment of space. This test assesses the satellite's ability to withstand temperature variations and the vacuum of space.
- *Electromagnetic Compatibility (EMC) Testing:* EMC testing ensures that the electronic components within the Cubesat do not interfere with each other and can operate without disruptions in the electromagnetic environment of space.

Satellite Set-up - Discussion

- *Power System Testing:* The power system, including solar panels and batteries, is thoroughly tested to verify its performance under various conditions.
- *Communication System Testing:* The communication system is tested to ensure reliable communication with ground stations. This involves assessing the performance of antennas, transceivers, and communication protocols.
- *Deployable Component Testing:* If the Cubesat includes deployable components such as antennas or solar panels, these are tested to ensure proper deployment in microgravity conditions.
- *Functional Testing:* All subsystems and components are tested together to verify their proper interaction and functionality as an integrated system. This includes testing the onboard computer, sensors, and any scientific instruments.
- *Software Testing:* The satellite's onboard software is rigorously test
- *Launch Vehicle Integration:* The Cubesat is integrated with the launch vehicle, and compatibility checks are performed to ensure a seamless integration process.

Satellite Set-up - Discussion

Fun video about the testing procedure of the HERA mission

<https://www.youtube.com/watch?v=mHAQILOdrR8>

BepiColombo vibration test

<https://www.youtube.com/watch?v=2jh3ViOpKqY>

Testing procedure of GALILEO

<https://www.youtube.com/watch?v=TfOPAuDgbiQ>

Satellite Set-up - Discussion

Testing on Cubesats is mandatory and must follow a specific sequence outlined by ECSS (European Cooperation for Space Standardization). Papers containing the test results are then submitted to the carrier (launch provider), and the satellite cannot be loaded onto the rocket without them.

Integrating a complete satellite is a time-consuming process, often taking months or even years due to complex technologies and extensive testing. Time is a critical factor in space missions, and securing a launch opportunity is essential. Launch dates are determined by the launch provider, and once you reserve a slot, you must submit your satellite by the specified date, even if launches get delayed.



Satellite Set-up - Discussion

If you successfully complete all the tests on your flight hardware, certain aspects beyond the company's control can nullify the time and effort you invested, keeping you from boarding the launch.

In the new space economy, especially in the Cubesat and small satellite industry, many startups and companies are accustomed to dealing with highly technological aspects. However, bureaucracy may move at a different speed than engineers, requiring advance planning for regulatory aspects.

For example, the satellite may be ready for placement on the rocket, but certain permissions, like the radio communication license, could be missing. Failure to obtain them on time could result in being denied the opportunity to board the launch.

Satellite Set-up - Discussion

Less common, though it occurs, is when the payload providers are unable to complete the hardware or software on time and, as a result, fail to integrate within the Cubesat ecosystem.

Various reasons could contribute to this, such as investor pressure for In-Orbit Demonstration (IOD), changes in plans, or upgrades.

Nonetheless, the consequence is a properly launched satellite with no data to collect, rendering it a flying object without utility.

No connection available

Satellite Design & Integration

END of DAY 3

